

DRONE MACHINE LEARNING

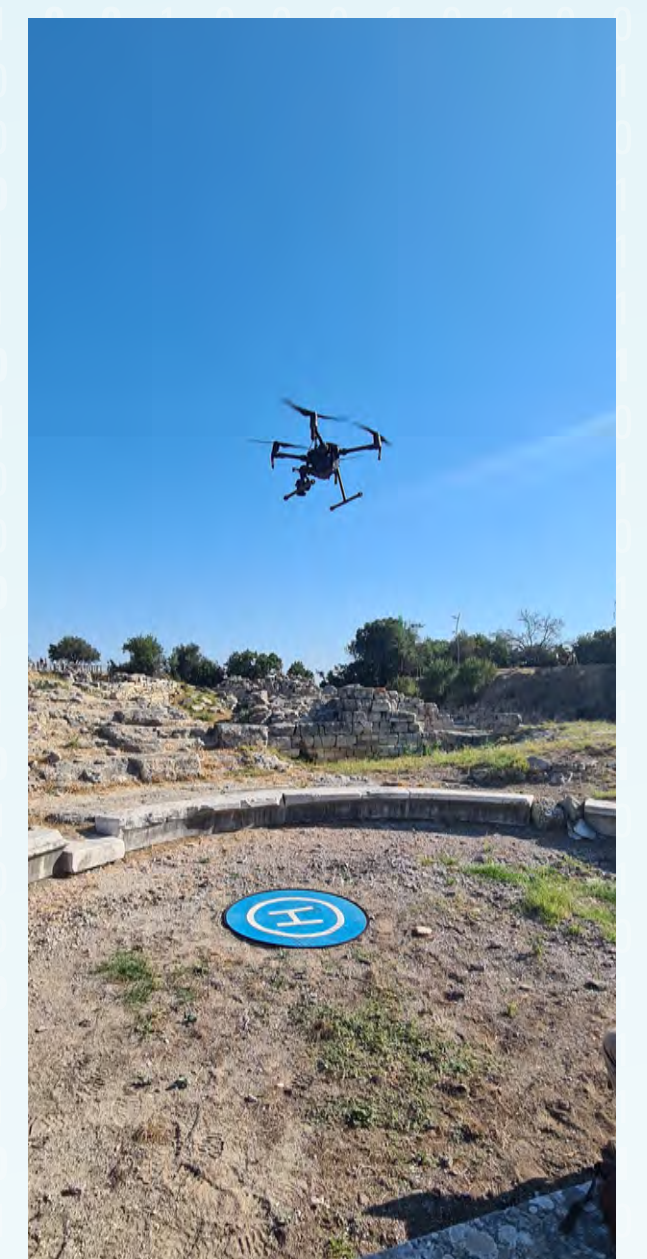
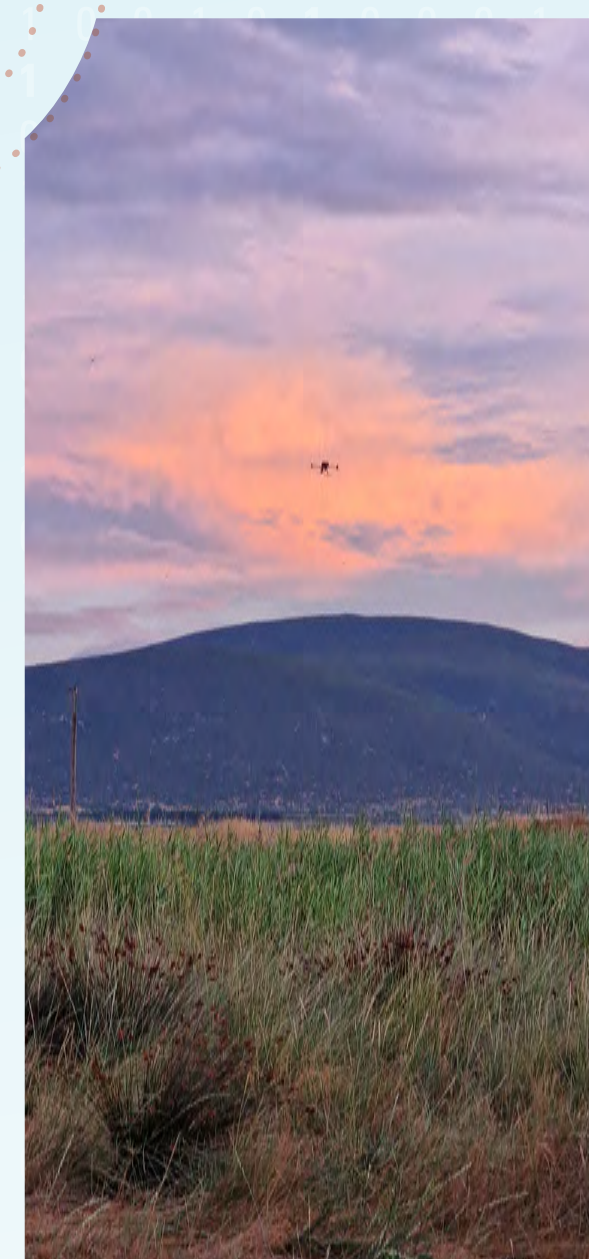
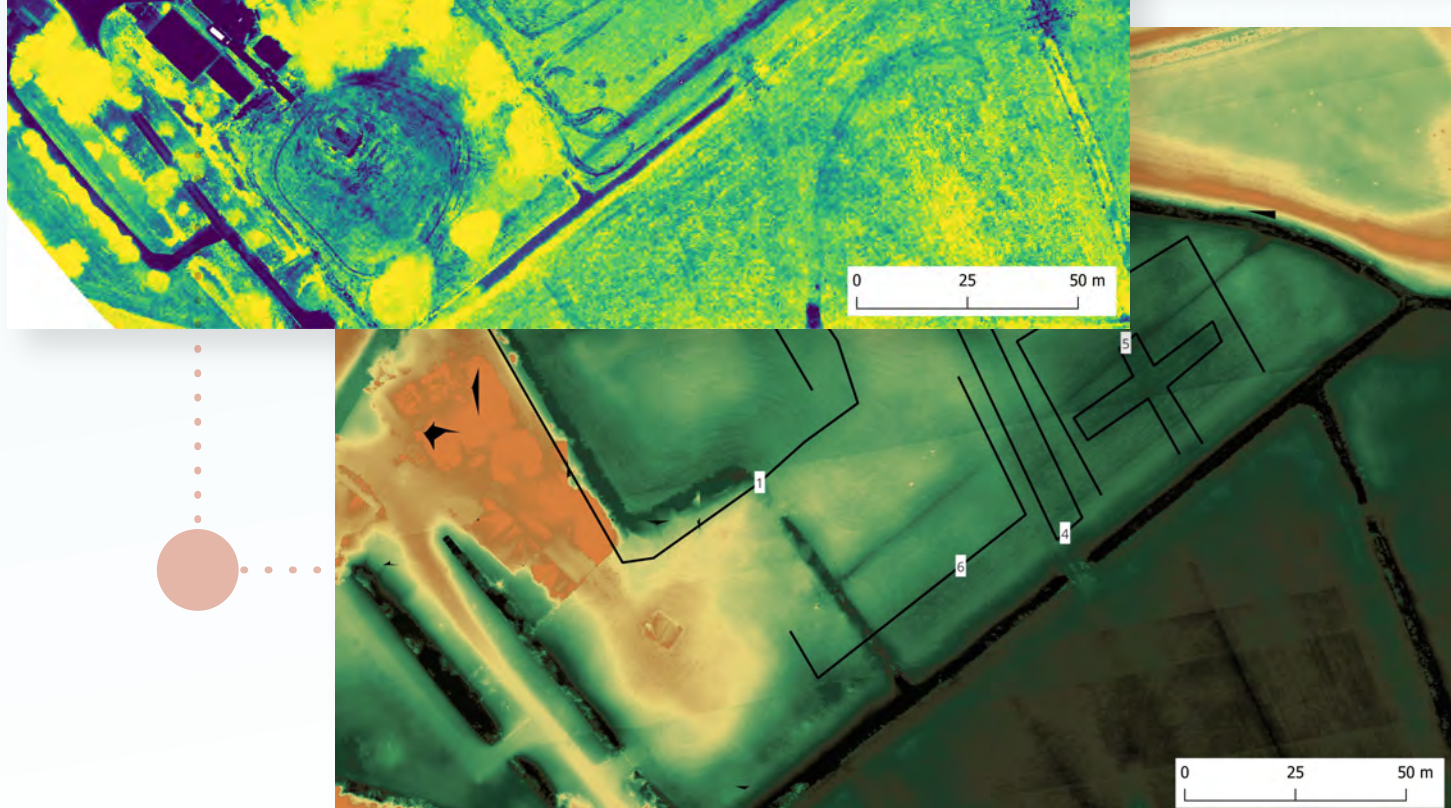
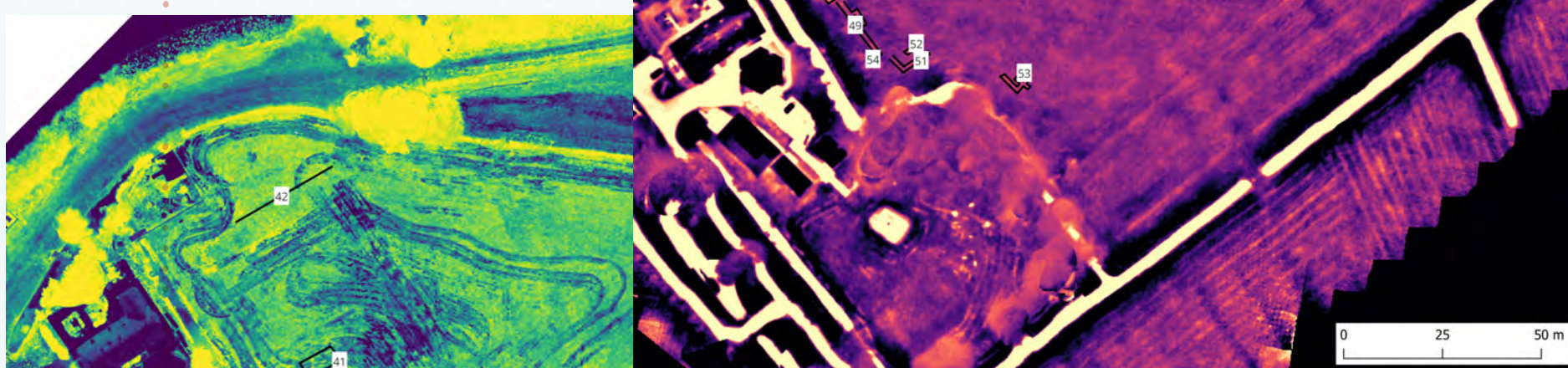
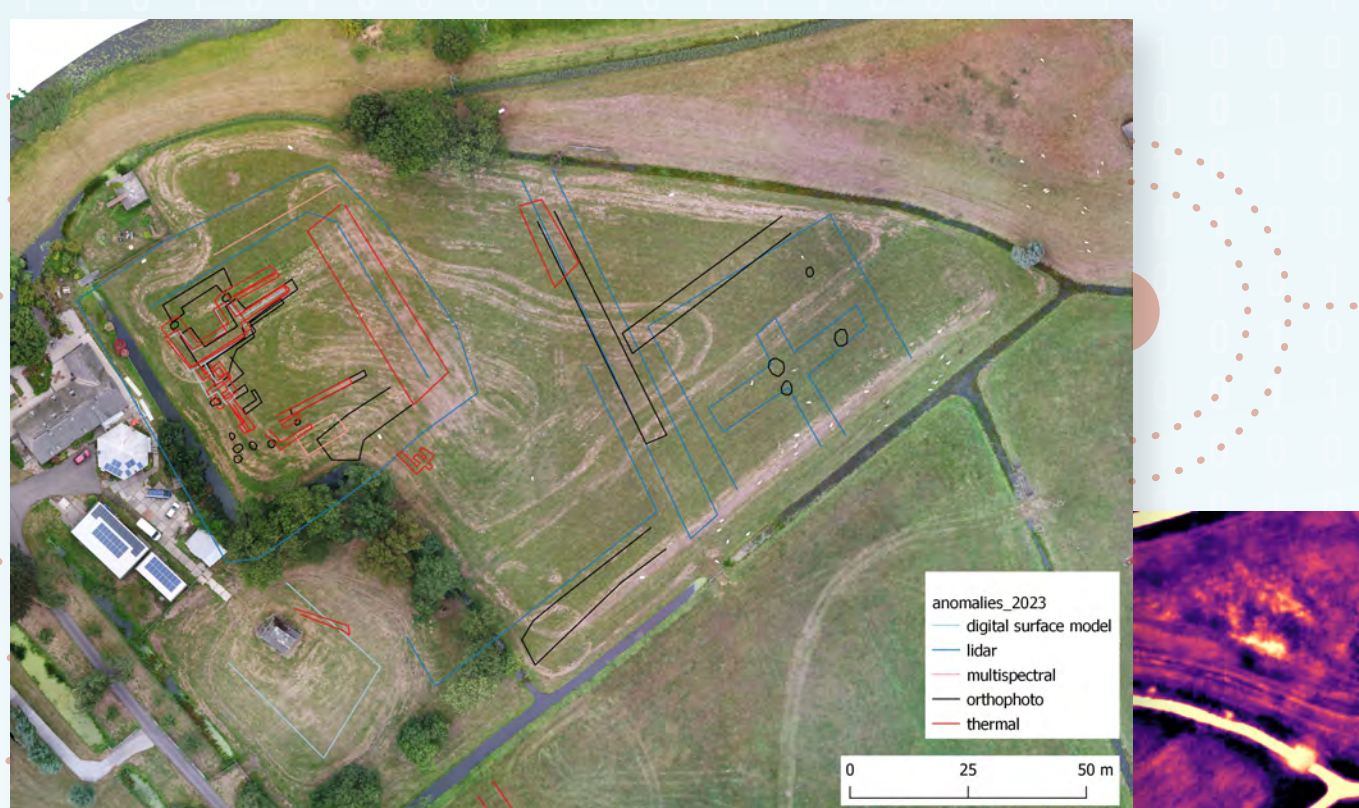
PROCESSING ARCHAEOLOGICAL SENSOR DATA FROM DRONES USING MACHINE LEARNING

Archaeological remote sensing is of paramount importance to face challenges of climate change, agricultural intensification, and landscape development that are threatening buried archaeological remains.

Drones using state-of-the-art sensor techniques strongly impact the possibilities of archaeologists by enabling them to trace ground and subsurface human activity in the past.



This innovative technology however also results in huge high-resolution and multimodal datasets. Therefore, there is an urgent need for computer-aided inspection of these complex datasets.



DroneML develops Open-Source software that aids the user to rapidly identify archaeological features of interest. The user will interactively provide some positive (e.g. traces of buried walls) and negative examples (e.g. bushes) in the remotely sensed drone captured data, i.e. high resolution optical scan, multispectral images, temperature mappings and elevation, after which a Machine Learning algorithm will provide suggestions for similar features in the data.

The research software that we develop is meant to facilitate the work of archaeologists, as well as widen research possibilities, both within the field of heritage as well as any other discipline making use of remotely sensed data capture techniques.

<https://research-software-directory.org/projects/droneml>